# The ideal step in the energy representation

D. Teaney[1]

[1]*Department of Physics and Astronomy, Stony Brook University, Stony Brook, New York 11794, USA*

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## Abstract

The Hamiltonian is written as the sum of the particle energies, $H = \sum_a e_a$, each evaluated from the particle's own entropy, momentum and volume. The volume is coarse grained with the entropy weights already in the code, $V_a = s_a/S(\boldsymbol{x}_a)$. Hamilton's equations then give an ideal step in which each particle moves with its own velocity and feels only a pressure force. Viscosity is left out.

## I. VARIABLES AND THE VOLUME

Each particle $a$ has a position $\boldsymbol{x}_a$, a momentum $\boldsymbol{m}_a$ and an entropy weight $s_a$. The weights are constants of the ideal step; a convenient start is $s_a = 1$ for all particles. The coarse-grained entropy density at the particle is the kernel sum

$$S_a \equiv S(\boldsymbol{x}_a) = \sum_b s_b\, W(\boldsymbol{x}_a - \boldsymbol{x}_b), \tag{1}$$

and the particle's volume is defined by

$$V_a = \frac{s_a}{S_a}, \qquad \text{so that} \qquad \frac{1}{V_a} = \frac{S_a}{s_a} \xrightarrow{s_a=1} \sum_b W(\boldsymbol{x}_a - \boldsymbol{x}_b). \tag{2}$$

For unit weights $1/V_a$ is the number density of particles. The particle's entropy density $s_a/V_a$ is then the smoothed one, $S_a$. Its momentum density is its own, $\boldsymbol{m}_a/V_a$, not smoothed.

When the particles move, the volume changes as

$$\frac{\dot V_a}{V_a} = -\frac{\dot S_a}{S_a} = \frac{1}{S_a}\sum_b s_b\,(\dot{\boldsymbol{x}}_b - \dot{\boldsymbol{x}}_a)\cdot \boldsymbol{D}_{ab}, \qquad \boldsymbol{D}_{ab} \equiv \frac{\partial W(\boldsymbol{x}_a - \boldsymbol{x}_b)}{\partial \boldsymbol{x}_a}, \tag{3}$$

the SPH estimate of $\nabla\cdot\dot{\boldsymbol{x}}$ at $\boldsymbol{x}_a$.

## II. THE PARTICLE ENERGY

The equation of state gives the energy density $h(S, \boldsymbol{M})$ as a function of the entropy and momentum densities. The particle's energy is

$$e_a = E_a(s_a, \boldsymbol{m}_a, V_a) \equiv V_a\, h\Big(\frac{s_a}{V_a}, \frac{\boldsymbol{m}_a}{V_a}\Big), \tag{4}$$

the same particle energy as in the particle representation of `changerep.tex`. Its differential is the first law for the particle,

$$de_a = T_a\, ds_a + \boldsymbol{v}_a\cdot d\boldsymbol{m}_a - p_a\, dV_a, \tag{5}$$

where $T_a$ (bare, $T_{\rm rel}/\gamma$), $\boldsymbol{v}_a$ and $p_a$ are the temperature, velocity and pressure at the particle's densities $(S_a, \boldsymbol{m}_a/V_a)$.

## III. THE HAMILTONIAN AND HAMILTON'S EQUATIONS

The Hamiltonian is the sum of the particle energies,

$$\boxed{H(\boldsymbol{x},\boldsymbol{m}) = \sum_a e_a = \sum_a V_a\, h\Big(\frac{s_a}{V_a}, \frac{\boldsymbol{m}_a}{V_a}\Big), \qquad V_a = \frac{s_a}{S(\boldsymbol{x}_a)}} \tag{6}$$

It depends on the positions only through the volumes. Since $s_a = V_a S_a$, eq. (6) is the same as

$$H = \sum_a s_a\, \mathcal{E}\Big(S_a, \frac{\boldsymbol{m}_a}{s_a}\Big), \tag{7}$$

with $\mathcal{E}$ the energy per unit entropy. The present Hamiltonian (`sph2.tex`) is $\sum_a s_a \mathcal{E}(S_a, \boldsymbol{M}_a/S_a)$, with the smoothed momentum $\boldsymbol{M}_a = \sum_b \boldsymbol{m}_b W_{ab}$ in place of the particle's own $\boldsymbol{m}_a$; that is the only difference.

Hamilton's equations are

$$\dot{\boldsymbol{x}}_a = \frac{\partial H}{\partial \boldsymbol{m}_a}, \qquad \dot{\boldsymbol{m}}_a = -\frac{\partial H}{\partial \boldsymbol{x}_a}, \qquad \dot{s}_a = 0. \tag{8}$$

By eq. (5), $\partial H/\partial \boldsymbol{m}_a = \boldsymbol{v}_a$: the particle moves with its own velocity,

$$\boxed{\dot{\boldsymbol{x}}_a = \boldsymbol{v}_a} \tag{9}$$

For the force, $\partial H/\partial \boldsymbol{x}_a = -\sum_c p_c\, \partial V_c/\partial \boldsymbol{x}_a$ with $\partial V_c/\partial \boldsymbol{x}_a = -(V_c/S_c)\, \partial S_c/\partial \boldsymbol{x}_a$, and

$$\frac{\partial S_c}{\partial \boldsymbol{x}_a} = \delta_{ca} \sum_b s_b \boldsymbol{D}_{ab} + s_a \boldsymbol{D}_{ac}. \tag{10}$$

Collecting terms,

$$\boxed{\dot{\boldsymbol{m}}_a = -\sum_b s_a s_b \left(\frac{p_a}{S_a^2} + \frac{p_b}{S_b^2}\right) \boldsymbol{D}_{ab}} \tag{11}$$

This is the standard symmetric SPH pressure force, with $S$ in the role of the mass density and $s_a$ in the role of the mass. Compared with `sph2.tex`, the terms in $\Delta\mathcal{M}_b(\boldsymbol{x}_a) = \boldsymbol{m}_b/s_b - \boldsymbol{M}_a/S_a$ are gone, because the momentum is not smoothed.

*a. Continuum limit.* With $\sum_b V_b f_b \boldsymbol{D}_{ab} \to \nabla f$, eq. (11) becomes $\dot{\boldsymbol{m}}_a \to -V_a \nabla p$. This is the momentum equation for a parcel moving with $\boldsymbol{v}$: relative to such a parcel the momentum flux $T^{ij} = M^i v^j + p\,\delta^{ij}$ reduces to $p\,\delta^{ij}$.

*b. The particle energies.* From eq. (5) at fixed $s_a$,

$$\dot{e}_a = \boldsymbol{v}_a \cdot \dot{\boldsymbol{m}}_a - p_a \dot{V}_a, \tag{12}$$

with $\dot{V}_a$ from eq. (3). The particle energies change by the work done on the particle; their sum does not.

## IV. WHAT IS CONSERVED

1. *Energy.* $H = \sum_a e_a$ is conserved by Hamilton's equations, and it is the same sum of particle energies eq. (4) that the viscous step of `changerep.tex` works with: the smoothed and the particle energies are one and the same.

2. *Momentum.* $\boldsymbol{D}_{ba} = -\boldsymbol{D}_{ab}$, so the pair forces in eq. (11) cancel and $\sum_a \boldsymbol{m}_a$ is conserved.

3. *Entropy.* Each $s_a$ is constant, so $\sum_a s_a$ is conserved, and each particle's entropy $S_a(e_a, \boldsymbol{m}_a)$ at volume $V_a$ equals $s_a$.

## V. WHAT IT DOES NOT CHANGE: THE STATIC RIGIDITY

At rest, $\boldsymbol{m}_a = 0$, the new and the present Hamiltonians are the same function of the positions,

$$H\big|_{\boldsymbol{m}=0} = \sum_a s_a\, \mathcal{E}(S_a, 0), \tag{13}$$

because the two differ only in which momentum enters $\mathcal{E}$. The static phonon spectrum of `rigidity.tex` (Sect. "The ideal Hamiltonian's phonons"), and with it the static shear modulus $G$ and $c_t$, are therefore unchanged: the positions still enter only through the kernel sum $S_a = \sum_b s_b W_{ab}$, which is what breaks the relabeling symmetry. The energy representation is *not* a fix for the rigidity. (Volumes that partition space, $\sum_a V_a = A$, would be; that is a different choice of $V_a$ in eq. (2), and eq. (6)–eq. (11) would carry over with $\partial V_c/\partial \boldsymbol{x}_a$ from the partition.)

What does change is the dynamics: the particles move with their own velocities rather than a doubly smoothed one, and the force has no $\Delta\mathcal{M}$ terms. Whether that changes the measured transverse oscillation in $C_T$ (its frequency or damping) was left to the runs. It does: in the full stochastic scheme the oscillation is gone (Sect. XI), although the static $c_t$ is the same.

## VI. THE DETERMINISTIC VISCOUS STEP

With $H = \sum_a e_a$, the viscous step is done on the particles' own variables; there is no change of representation. The derivative of $H$ with respect to a weight at fixed $\boldsymbol{x}$, $\boldsymbol{m}$ is

$$T_a^{\text{eff}} \equiv \frac{\partial H}{\partial s_a} = T_a - \frac{p_a}{S_a} + \sum_b s_b \frac{p_b}{S_b^2} W_{ab} = T_a + O(h^2), \tag{14}$$

using $\mathcal{E} - \boldsymbol{v}\cdot\boldsymbol{m}/s = T - p/S$; the last two terms are the smoothed minus the local $p/S$. With

$$\boldsymbol{\beta}_a = \frac{\boldsymbol{v}_a}{T_a^{\text{eff}}}, \qquad \Delta\boldsymbol{m}_a = -\Delta t\, \frac{\partial K}{\partial \boldsymbol{\beta}_a}, \qquad \Delta s_a = -\boldsymbol{\beta}_a \cdot \Delta\boldsymbol{m}_a, \tag{15}$$

$\Delta H = \sum_a (T_a^{\text{eff}} \Delta s_a + \boldsymbol{v}_a \cdot \Delta\boldsymbol{m}_a) = 0$ to first order in $\Delta t$, and $\sum_a \Delta s_a = 2K\Delta t \geq 0$, as in `sph2.tex`. $K$ is the bond dissipative function of `changerep.tex`, which already takes a pointwise $\boldsymbol{\beta}_a$.

Compared with `changerep.tex`: the forward map $s_a \to e_a$ is no longer a map, since $e_a$ is already the energy in $H$; the gradient correction $g_a$ (its Appendix A) was the difference

between $\sum_a e_a$ and the smoothed $H$, and is now zero. The exact back map at fixed $e_a$, $V_a$ does not carry over as it is: a change of $s_a$ changes $S_b$, hence $V_b$, at every neighbour, so fixing every $e_a$ at fixed $V_a$ does not fix $H$ exactly. The step eq. (15) conserves $H$ to first order in $\Delta t$; the drift is measured (plan, step 6).

## VII. IMPLEMENTATION PLAN

The Hamiltonian is chosen by a new field `params.hamiltonian` $\in$ {`:smoothed`, `:energy`} (environment `SPH_HAMILTONIAN`), default `:smoothed`, so that every existing run and test is unchanged.

*a. Unchanged.* The `Particle` struct (`x`, `s`, `m`, `S_smooth` and the EOS fields are enough); kernels, cell and neighbour lists; the EOS solvers `eos_newton`, `solve_eos!`, `particle_energy`, `eos_from_energy`; the integrators `leapfrog_step!` and `kdk_step!` (their structure; they call the functions below); the bond dissipative function and force; the set-up of initial conditions ($s_a = S(\boldsymbol{x}_a)\,\mathrm{vol}$, or $s_a = 1$); the recorders. The stochastic step is left alone.

*b. Changed (all in `celllist.jl`, dispatched on `params.hamiltonian`).*

1. `smooth_to_particles!`: sum $S_a$ only, and set the momentum density to the particle's own, `M_smooth` $= \boldsymbol{m}_a S_a/s_a = \boldsymbol{m}_a/V_a$. Then `solve_eos!` returns, unchanged, the particle's own $T_a$, $\boldsymbol{v}_a$, $p_a$ and $e_a/s_a$ (`H_over_S`). One neighbour sum fewer.

2. `update_velocities!`: $\dot{\boldsymbol{x}}_a = \boldsymbol{v}_a$, local, no neighbour sum; `Teff` from eq. (14), one neighbour sum (needed only by the viscous step and by the bond metric).

3. `hamiltonian_force!`: the pair bracket becomes $-s_a s_b(p_a/S_a^2 + p_b/S_b^2)$, eq. (11).

4. Viscous step: `dissipative_step_changerep!` with $\boldsymbol{\beta}_a$ of eq. (15) read from the particle (no `pointwise_state` copy), and $\Delta s_a = -\boldsymbol{\beta}_a \cdot \Delta \boldsymbol{m}_a$. `dissipative_step!` calls it, not the gradient-corrected step, when `params.hamiltonian == :energy`.

5. Diagnostics: the total energy is $\sum_a e_a = \sum_a s_a\,\mathtt{H_over_S}_a$, the same number as $H$.

*c. Steps and checks.*

1. Implement 1–3 behind the flag; `test/testregress_step.jl` must stay bit-identical for `:smoothed`.

2. A test `test/testenergystep.jl`: $\partial H/\partial \boldsymbol{m}_a$ and $\partial H/\partial \boldsymbol{x}_a$ against finite differences of $H$ on a jittered lattice; $\sum_a \dot{\boldsymbol{m}}_a = 0$; $\dot{s}_a = 0$.

3. Ideal runs: energy drift of $H$ versus $\Delta t$ (second order for the leapfrog), a sound wave (speed and damping) against the present step (`examples/testsound.jl`), and the time step: the unsmoothed $\boldsymbol{v}_a$ may need a smaller CFL.

4. Static check: the phonon spectrum of Sect. V must agree with the present one to roundoff (it is the same $H$ at $\boldsymbol{m} = 0$).

5. Implement 4; check $\sum_a \Delta s_a = 2K\Delta t$ and $\Delta H = O(\Delta t^2)$ per step; a shear wave against $e^{-\nu k^2 t}$ (`examples/testshearwave.jl`).

6. Energy drift over long coupled deterministic runs, against the present step with the gradient-corrected map.

*d. Documentation.* Each changed function gets a comment that states the equation it implements and cites this note by equation; `params.hamiltonian` is described where `SimParams` is defined; the outcome of each check above is added to this note as it is done.

*e. Status (2026-11-09).* Items 1–3 of the changes and item 4 (the viscous step, `dissipative_step_energy!`) are in `src/sph.jl` and `src/celllist.jl`. On a jittered lattice with a moving fluid, $\partial H/\partial \boldsymbol{m}_a$, $\partial H/\partial \boldsymbol{x}_a$ and $\partial H/\partial s_a$ agree with $\dot{\boldsymbol{x}}_a$, $-\dot{\boldsymbol{m}}_a$ and $T_a^{\mathrm{eff}}$ to $10^{-6}$ (finite differences); $\sum_a \dot{\boldsymbol{m}}_a = 7 \times 10^{-16}$; $H = \sum_a$`particle_energy` exactly. The deterministic shear wave is Sect. VIII.

## VIII. THE DETERMINISTIC SHEAR WAVE

`examples/shearwave_decay.jl` (viscous and ideal) and `examples/shearwave_scan.jl` (ideal, kernels and $h/a$). A uniform fluid with

$$M_y(x,0) = A \sin kx, \qquad k = \frac{2\pi}{L}, \tag{16}$$

evolved with the full deterministic step. A fluid keeps the amplitude with $\eta = 0$ and decays as $e^{-\nu k^2 t}$, $\nu = (\eta/s)/T_{\mathrm{rel}}$, with $\eta > 0$. An elastic solid with transverse speed $c_t$ instead gives, with $\eta = 0$,

$$\frac{A(t)}{A(0)} = \cos(c_t k t). \tag{17}$$

At $h/a = 2$ (C2, $kh = 0.31$):

1. The two Hamiltonians agree: amplitudes to $10^{-3}$ with and without viscosity, $|\Delta H/H| \sim 10^{-11}$ for the ideal step. The viscous step drifts by $4 \times 10^{-8}$ (`:energy`) and $8 \times 10^{-9}$ (`:smoothed`).

2. For $A \le 0.005$ (displacements $\le 0.4a$) the ideal wave is an undamped cosine with $c_t = 0.0340$ (`:energy`) and 0.0335 (`:smoothed`): the static value 0.034 of `rigidity.tex`. For $A = 0.02$ the particles slip by $2.4a$ and the response is not linear.

3. With $\eta/s = 0.02$ the amplitude falls below $e^{-\nu k^2 t}$ and turns negative: a damped transverse oscillation.

*a. Kernels and $h/a$.* `examples/shearwave_scan.jl`: ideal step only, $A = 0.002$, a periodic strip $L_x = 1$, $L_y \simeq 6h$ (2D, the wave depends on $x$ only; 560 to 3400 particles), $t \leq 30$. $c_t$ is a least-squares fit of $\cos(c_t k t)$, or $\cosh(|c_t| k t)$ for an unstable wave (written $c_t < 0$), over the first quarter period. Dynamic $c_t$ (`:energy` / `:smoothed`) against the static phonon values of `rigidity.tex` ($qh \simeq 0.26$; here $kh = 0.31$):

| kernel | | $h/a = 2$ | 2.5 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| C2 | dynamic | 0.0340/0.0335 | 0.0308/0.0304 | 0.0214/0.0211 | $-0.005$ [a] | $-0.0027/-0.0027$ |
| | static | 0.035 | 0.031 | 0.022 | $-0.011$ | $-0.0034$ |
| C4 | dynamic | grows [b] | $-0.0047/-0.0047$ | $-0.0018/-0.0018$ | 0.0022/0.0022 | 0.0008/0.0008 |
| | static | $-0.0054$ | unstable | $-0.0029$ | 0.0029 | 0.0008 |
| C6 | dynamic | 0.0170/0.0169 | 0.0065/0.0064 | 0.0027/0.0026 | 0.0004/0.0004 | 0.0002/0.0002 |
| | static | 0.017 | 0.0066 | 0.0026 | $-0.0003$ | 0.0002 |

[a] grows, then not linear (poor fit). [b] $A$ rises to 1.03 by $t \simeq 10$, then the displacement reaches $0.9a$ and the response is not linear.

1. The dynamic and static $c_t$ agree, signs included, wherever the wave stays linear: the transverse sound of the runs is the elastic stiffness of $H$, and the time stepping adds nothing.

2. The two Hamiltonians agree to 1–2%, as they must ($H$ is the same function of the positions at $\boldsymbol{m} = 0$). The energy representation by itself does not change the rigidity.

3. $|c_t|$ falls with $h/a$ and with the kernel order, oscillating in sign (C2, C4); C6 is below $10^{-3}$ from $h/a = 4$ on. $\Delta H/H \leq 10^{-13}$ in every linear run.

(The strips at $h/a = 3$ and 5 have two cell rows in $y$; they first blew up through a double counting in the cell list for `ncy` = 2, fixed in `neighbor_range`.)

*b. The deterministic test suite.* `examples/test*.jl` with `SPH_HAMILTONIAN=smoothed` and `energy` (one thread; PyPlot does not run in a multithreaded process). Residuals are rms against the analytic solution:

| test | | `:smoothed` | `:energy` |
|---|---|---|---|
| Gaussian 2D, $t = 2.2$ | $\Delta S/S$ | 0.0362 | 0.0363 |
| | $\Delta H/H$ | $1.9 \times 10^{-5}$ | $2.4 \times 10^{-5}$ |
| Gaussian 1D, $t = 2.2$ | $\Delta S/S$ | 0.0512 | 0.0513 |
| | $\Delta H/H$ | $1.4 \times 10^{-4}$ | $1.8 \times 10^{-4}$ |
| viscous sound 2D (amplitude $2.5 \times 10^{-3}$) | residual | $1.9 \times 10^{-4}$ | $3.4 \times 10^{-5}$ |
| | $\Delta H/H$ | $1.1 \times 10^{-8}$ | $4.0 \times 10^{-8}$ |
| viscous sound 1D (amplitude $5.2 \times 10^{-3}$) | residual | $2.2 \times 10^{-5}$ | $1.5 \times 10^{-5}$ |
| | $\Delta H/H$ | $7 \times 10^{-10}$ | $2.5 \times 10^{-9}$ |
| ideal sound, one period (amplitude $5.2 \times 10^{-3}$) | residual | $5.6 \times 10^{-3}$ | $1.8 \times 10^{-3}$ |
| shear diffusion (amplitude $1.0 \times 10^{-3}$) | residual | $1.55 \times 10^{-5}$ | $1.24 \times 10^{-5}$ |

The entropy production agrees to 0.1%; `:energy` follows the analytic waves as well or better (no doubly smoothed velocity); its energy drift is larger by 1.3–4, from the viscous step, which conserves $H$ to first order in $\Delta t$.

## IX. PROPOSAL: VOLUMES THAT ADD UP TO THE AREA

*a. Why the kernel volume gives a modulus.* Since $H$ depends on the positions only through the volumes, to second order in their changes

$$\delta H = -\sum_a p_a\,\delta V_a + \sum_a \frac{K_a}{2V_a}\,\delta V_a^2, \tag{18}$$

with $K$ the bulk modulus.

*b. The bulk modulus of a moving particle.* At fixed $s_a$ and $\boldsymbol{m}_a$ a change of $V_a$ scales the densities along a ray, $S \to \lambda S$, $\boldsymbol{M} \to \lambda \boldsymbol{M}$, $\lambda = V_0/V$: a line of constant $\boldsymbol{M}/S$, the adiabat of `theory.tex`. Along it $d\varepsilon = (TS + \boldsymbol{v}\cdot\boldsymbol{M})\,d\lambda/\lambda = (\varepsilon + p)\,d\lambda/\lambda$, with $\varepsilon = T^{00}$, so

$$\delta T^{0\nu} = \frac{\delta\lambda}{\lambda}\,w\gamma\,u^\nu, \tag{19}$$

and

$$\frac{\partial e_a}{\partial V_a} = -p_a, \qquad K_a \equiv V_a\frac{\partial^2 e_a}{\partial V_a^2} = (\varepsilon+p)\Big(\frac{dp}{d\varepsilon}\Big)_{\boldsymbol{M}/S} = \frac{w\,c_s^2}{1-c_s^2v^2}. \tag{20}$$

In terms of the inverse susceptibility $\chi^{-1}_{\mu\nu} = \partial\beta_\mu/\partial T^{0\nu}$ of `theory.tex` (eq. `inversesusc`),

$$K_a = \frac{w^2\gamma}{\beta}\,u^\mu\chi^{-1}_{\mu\nu}u^\nu, \tag{21}$$

since $u^\mu(u_\mu + \Delta^0_\mu/\gamma) = -1$ and $u^\mu\Delta_{\mu\nu} = 0$ leave only the adiabat term. At rest $K = w\,c_s^2 = s\,dp/ds$, the adiabatic bulk modulus; in a moving fluid it is larger by $1/(1-c_s^2v^2)$, the factor of transverse sound in `theory.tex`. To second order the position dependence of $H$ is thus set by $p$ and by the adiabat projection of $\chi^{-1}$. Checked against finite differences of the particle energy (conformal EOS, $v = 0$–$0.9$): $\partial e/\partial V = -p$ and eq. (20) to $10^{-6}$.

For $V_a = s_a/S_a$ a homogeneous shear $\epsilon$ of a lattice changes every volume by the same amount at second order, $\delta V_a = c\,\epsilon^2$, so $\sum_a \delta V_a \neq 0$, and the first term gives a modulus

$$G \sim p\,c(h/a, \text{kernel}). \tag{22}$$

This is the $\epsilon$ of `rigidity.tex`, proportional to the pressure.

*c. The condition.* If the volumes always add up to the area,

$$\sum_a V_a = A, \tag{23}$$

then at uniform pressure $\sum_a p\,\delta V_a = p\sum_a \delta V_a = 0$, and the second term of eq. (18) is of fourth order in $\epsilon$ on a lattice: no modulus. A partition of space (Voronoi cells, or the partition of unity of meshless finite-mass methods) satisfies eq. (23); so does the simplest choice, a global normalization of the kernel volume,

$$V_a = \frac{A}{Z}\,\tilde V_a, \qquad \tilde V_a = \frac{s_a}{S_a}, \qquad Z = \sum_c \tilde V_c. \tag{24}$$

The particle's entropy density is then $s_a/V_a = (Z/A)\,S_a$.

*d. Hamilton's equations.* With eq. (24) in eq. (6), $\partial H/\partial \boldsymbol{x}_a = -\sum_c p_c\, \partial V_c/\partial \boldsymbol{x}_a$ and

$$\frac{\partial V_c}{\partial \boldsymbol{x}_a} = \frac{A}{Z}\Big(\frac{\partial \tilde{V}_c}{\partial \boldsymbol{x}_a} - \frac{\tilde{V}_c}{Z}\sum_d \frac{\partial \tilde{V}_d}{\partial \boldsymbol{x}_a}\Big). \tag{25}$$

The second term subtracts the volume-averaged pressure

$$\bar{p} = \frac{1}{A}\sum_c p_c V_c, \tag{26}$$

and the force is eq. (11) with $p \to p - \bar{p}$:

$$\dot{\boldsymbol{m}}_a = -\frac{A}{Z}\sum_b s_a s_b \left(\frac{p_a - \bar{p}}{S_a^2} + \frac{p_b - \bar{p}}{S_b^2}\right) \boldsymbol{D}_{ab}. \tag{27}$$

The same substitution in eq. (14) gives the derivative with respect to the weights,

$$T_a^{\text{eff}} = T_a - \frac{A}{Z}\left[\frac{p_a - \bar{p}}{S_a} - \sum_b s_b \frac{p_b - \bar{p}}{S_b^2} W_{ab}\right]. \tag{28}$$

Still exact: $H$ is conserved by Hamilton's equations, and the pair terms of eq. (27) are odd under $a \leftrightarrow b$, so momentum is conserved. The continuum limit is unchanged, $\nabla(p-\bar{p}) = \nabla p$. The cost is one global sum per step. In SPH, subtracting a background pressure is a known remedy for the tensile instability; here it follows from the Hamiltonian, and energy conservation is kept.

*e. Expected, and tests.*

1. Perfect lattice: $c_t$ falls from 0.034 to near zero. Test: the shear-wave scan with eq. (24), against Sect. VIII.

2. Thermal state: what is left is driven by the pressure fluctuations $\delta p$ rather than by $p$, with random sign; no coherent long-wavelength modulus is expected. Test: the stochastic runs, Sect. X.

3. Disordered arrangements: an affine shear changes the individual $\delta V_a$ at first order, with random sign, which leaves a modulus $\propto K\,(\text{disorder})^2$ from the second term of eq. (18): the caveat (a) of `rigidity.tex` for Voronoi volumes. A local partition does not remove it either. Test: the static phonon spectrum on jittered lattices (`ideal_lattice_modes.jl`, `JITTER`).

Implementation: an option `volume = :normalized` of the `:energy` Hamiltonian; one reduction for $\bar{p}$ and $Z$, and $p \to p - \bar{p}$, $S \to (Z/A)S$ where the particle state is evaluated.

## X. THE STOCHASTIC STEP WITH `:ENERGY`

Not yet tested. The stochastic step (`stochrep.tex`) works at fixed positions, volumes $V_a = s_a/S_a$ and particle energies $e_a$, which are exactly the variables of eq. (6); on a frozen lattice it is the same step for both Hamiltonians. Two differences:

1. *No gradient correction.* The gradient energy $g_a$ of `changerep.tex` (Appendix A) absorbs the difference between $\sum_a e_a$ and the smoothed $H$, which is now zero. With `:energy` the step must use the plain back map $s'_a = S_a(e_a, \boldsymbol{m}'_a)$ (`gradient = false`).

2. *The back map and the volumes.* The back map holds $e_a$ and $V_a$ fixed, but a change of $s_a$ changes $S_b$ and so $V_b$ at every neighbour. To first order

$$\Delta H \simeq \sum_a \left(T_a^{\text{eff}} - T_a\right)\Delta s_a, \tag{29}$$

with $T_a^{\text{eff}} - T_a$ the smoothed minus the local $p/S$, of order $h^2$. In the deterministic step this is $4 \times 10^{-8}$ (Sect. VIII); with noise the $\Delta s_a$ are larger and fluctuate, so its size has to be measured.

Tests, in order:

1. `stochrep_step!` defaults to `gradient = false` when `params.hamiltonian == :energy`, and the drivers keep that default.

2. One step on a thermal state: $\Delta H$ per stochastic step, `:energy` against `:smoothed` (with its gradient-corrected map).

3. Coupled runs, `stochrep_hamiltonian.jl`: $L = 0.6$, $h/a = 2$, $\eta/s = 0.2$, $S_0 = 10^6$, to $t \approx 50$, 2–4 realizations per Hamiltonian: $\Delta H/H(t)$, KE/eq, the lattice disorder. Local.

4. At NERSC, after 1–3: equilibrium $C_T$, $C_L$ with `:energy`, as campaign `second`; the node-hour estimate first.

*a. Status.* Test 1 is done; `:smoothed` stays bit-identical (`test/testregress_step.jl`). Test 2 (`examples/energystep_stoch_onestep.jl`, $L = 0.6$, $h/a = 2$, equilibrium start, 20 half steps $\Delta t/2 = 0.0043$ at fixed positions, 3 seeds):

| | `:smoothed` (gradient map) | `:energy` |
|---|---|---|
| $\Delta H/H$ per half step, rms | $3.4 \times 10^{-5}$ | $1.8 \times 10^{-7}$ |
| $\Delta H/H$ per half step, mean | $\pm 7 \times 10^{-6}$ (seed to seed) | $-1.8 \times 10^{-7}$ (every seed) |

With `:energy`, $\Delta H = \Delta \sum_a e_a$ exactly, and the change is 200 times smaller than with `:smoothed`, but systematic: a cooling of about $4 \times 10^{-5}$ per unit time. It is correlated ($-0.8$) with $\sum_a (T_a^{\text{eff}} - T_a)\Delta s_a$, but of the opposite sign and twice the size; not understood yet.

With Sect. IX, the same tests with `volume = :normalized` decide whether the transverse oscillation of the equilibrium $C_T$ is the pressure term of eq. (18).

## XI. STOCHASTIC RUNS AT NERSC

Tests 3 and 4, run from commit `934aa86` with `stochrep_equilibrium.jl`; everything as in campaign `rigidity` (C2, $h/a = 2$, $\eta/s = 0.2$, $S_0 = 10^6$, $\Delta t = 8.66 \times 10^{-3}$), only `SPH_HAMILTONIAN` differs. Record in `nersc/LOG.md`.

*a. Equilibration (debug job 61055621).* Box 0.6 ($N = 672$), $t = 200$, 4 seeds per run, both Hamiltonians side by side on one node, Fig. 1. Left: the full scheme from a cold start (all momenta zero). Right: the ideal step alone (no viscous step) from a thermal start. Purpose: does `:energy` run at all, and does it heat?

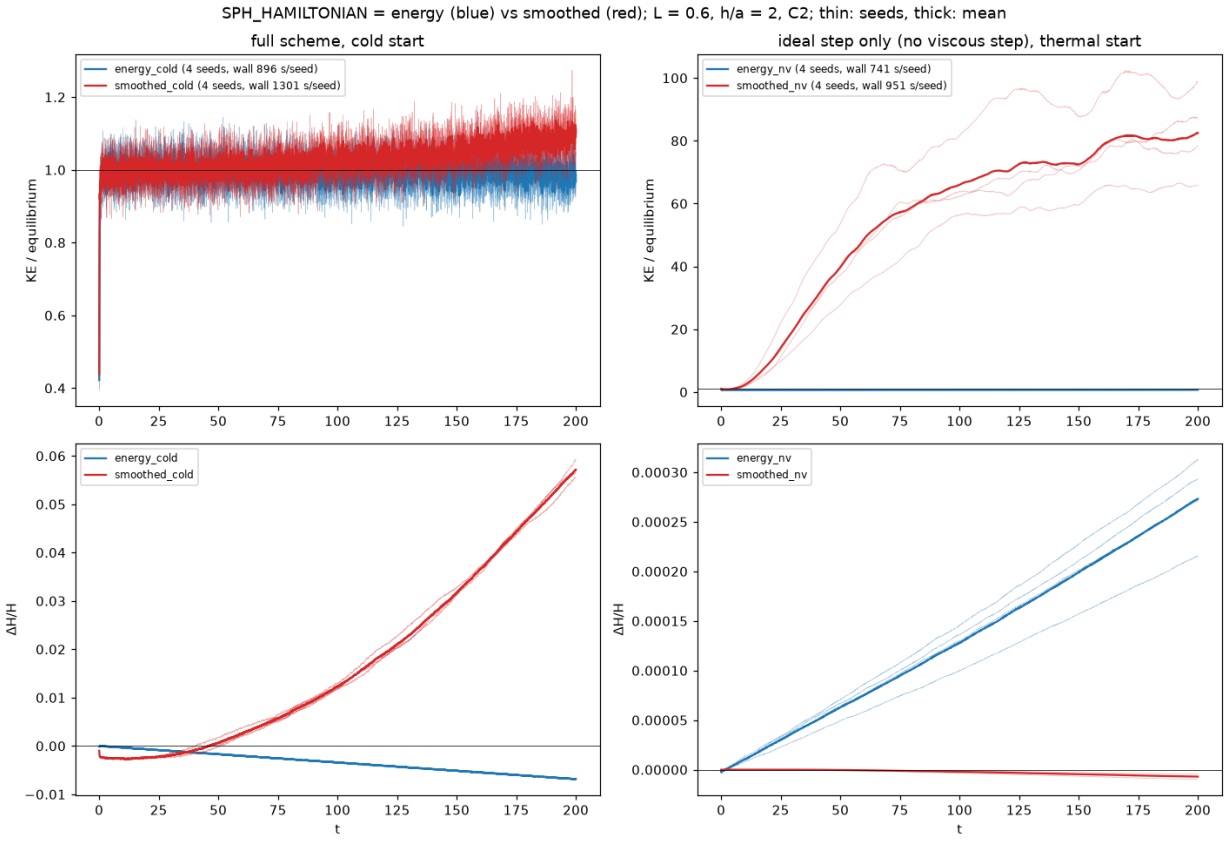



FIG. 1. KE divided by its equilibrium value (top) and $\Delta H/H$ (bottom) against $t$; blue `:energy`, red `:smoothed`; thin lines single seeds, thick the mean. Script `examples/energystep_equilibration.jl`.

1. Full scheme: both equilibrate within $t \approx 1$ and no step is rejected. `:smoothed` heats, faster than linearly ($\Delta H/H = +5.7 \times 10^{-2}$ at $t = 200$, KE/eq drifting to 1.05–1.10). `:energy` stays at KE/eq = 0.99 and cools linearly,

$$\frac{1}{H}\frac{dH}{dt} \simeq -3.4 \times 10^{-5}, \tag{30}$$

the same in every seed: the drift of test 2, now over $2 \times 10^4$ steps.

2. Ideal step alone: `:smoothed` conserves $H$ to $10^{-5}$ but its kinetic energy runs away, KE/eq = 60–95 at $t = 200$, taken from the internal energy. `:energy` settles at KE/eq $\approx 0.67$, the equipartition of a harmonic solid (half of the energy in the elastic modes), with $H$ drifting by $+1.4 \times 10^{-6}$ per unit time.

*b. Equilibrium correlators (job 61067905, resumed as 61073581).* Box 1.2 ($N = 2688$), equilibrium start, $t = 200$, 16 seeds, against the `:smoothed` set `ha2` of campaign `rigidity` (16 seeds), Fig. 2. The transverse correlator is fitted to the damped oscillator of `campaigns.tex`,

$$\frac{C_T(k,\tau)}{C_T(k,0)} = e^{-\gamma\tau}\Big[\cos\omega\tau - \frac{\gamma}{\omega}\sin\omega\tau\Big], \qquad \omega^2 = c_t^2k^2 - \gamma^2, \tag{31}$$

which is the fluid $e^{-\nu k^2\tau}$ for $c_t = 0$, $2\gamma = \nu k^2$.

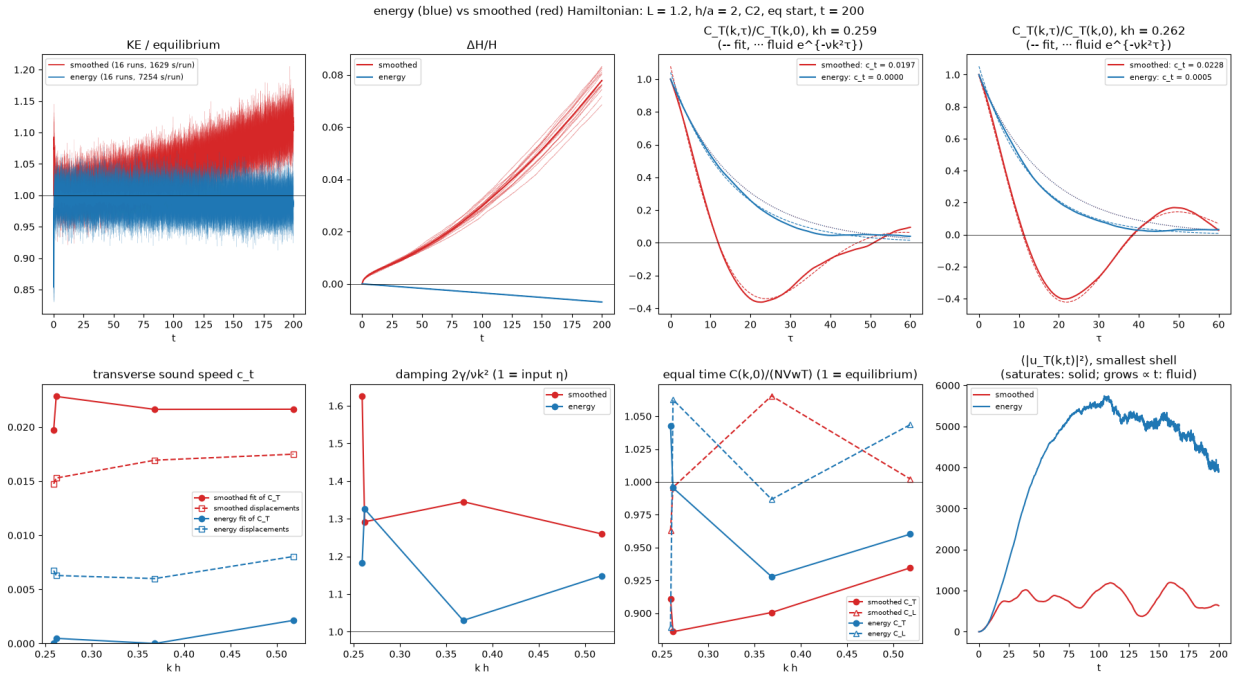



FIG. 2. `:energy` (blue) against `:smoothed` (red), $L = 1.2$, $h/a = 2$. Top: KE/eq and $\Delta H/H$ against $t$; $C_T(k,\tau)/C_T(k,0)$ in the two smallest shells with the fit (dashed) and the fluid $e^{-\nu k^2\tau}$ (dotted). Bottom: $c_t$ from the fit (circles) and from the displacements (squares, meaningful only for a solid); the damping $2\gamma/\nu k^2$; the equal-time $C_{T,L}(k,0)$; and $\langle|u_T(k,t)|^2\rangle$ in the smallest shell (displacements by minimum image in the box, so they wrap once the particles have wandered far). Script `examples/energystep_compare.jl`.

| | `:smoothed` | `:energy` |
|---|---|---|
| $c_t$ from the fit, $kh = 0.26$–$0.52$ | 0.020–0.023 | 0.000–0.002 |
| $2\gamma/\nu k^2$ | 1.26–1.63 | 1.03–1.33 |
| $C_T(k,0)/NV_awT$ | 0.89–0.93 | 0.93–1.04 |
| KE/eq | 1.05–1.10, rising | 0.99 |
| $\Delta H/H$ at $t = 200$ | $+7.8\times10^{-2}$ | $-6.9\times10^{-3}$ |
| wall time per seed (8 threads per 4 seeds) | $1.6\times10^3$ s | $7.3\times10^3$ s |

1. With `:energy` the transverse oscillation is gone: $C_T$ decays monotonically, close to $e^{-\nu k^2\tau}$, with a damping 0–30% above the input one, and the transverse displacements

grow far beyond the `:smoothed` plateau: the particles flow. The deficit $C_T(k, 0) < 1$ at small $k$ is mostly gone too.

2. This is not what Sect. V and the shear wave suggested, since the static $c_t$ is the same. A possible reason (not tested): with $\dot{\boldsymbol{x}}_a = \boldsymbol{v}_a$ neighbours can slip past one another, while the doubly smoothed velocity of `:smoothed` moves them together and keeps the arrangement locked. The ideal step alone is still a solid (KE/eq $\approx 0.67$ above); the flow needs the stochastic step.

3. The cooling of test 2 remains, at the same rate per unit time as in the small box.

4. At $L = 1.2$ `:energy` took 4.5 times longer per seed than `:smoothed` (which ran from the older commit `4663c3f`). The code is not slower: on one node, one seed with 8 threads to $t = 10$, it is 1.4 times faster (51 against 75 s). The slowdown builds up during the run; being measured.